

Multiversal views on language models

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Abstract

The virtuosity of language models like GPT-3 opens a new world of possibility for human-AI collaboration in writing. In this paper, we present a framework in which generative language models are conceptualized as multiverse generators. This framework also applies to human imagination and is core to how we read and write fiction. We call for exploration into this commonality through new forms of interfaces which allow humans to couple their imagination to AI to write, explore, and understand non-linear fiction. We discuss the early insights we have gained from actively pursuing this approach by developing and testing a novel multiversal GPT-3-assisted writing interface.

Keywords: writing assistant, hypertext narratives, multiverse writing, GPT-3

1 Introduction

GPT-3 [4], OpenAI’s new generative language model, has astonished with its ability to generate coherent, varied, and often beautiful continuations to natural language passages of any style. To creative writers and those who wish themselves writers, such a system opens a new world of possibilities. Some rightfully worry whether human writing will become deprecated or worthless in a world shared with such generative models, and others are excited for a renaissance in which the creative powers of human writers are raised to unprecedented heights in collaboration with AI. In order to achieve the latter outcome, we must figure out how to engineer human-machine interfaces that allow humans to couple their imaginations to machines and feel freed rather than replaced. We will present the still-evolving approach we have learned over several months of testing and designing interfaces for writing with the aid of GPT-3, beginning by introducing the framework of language models as multiverse generators.

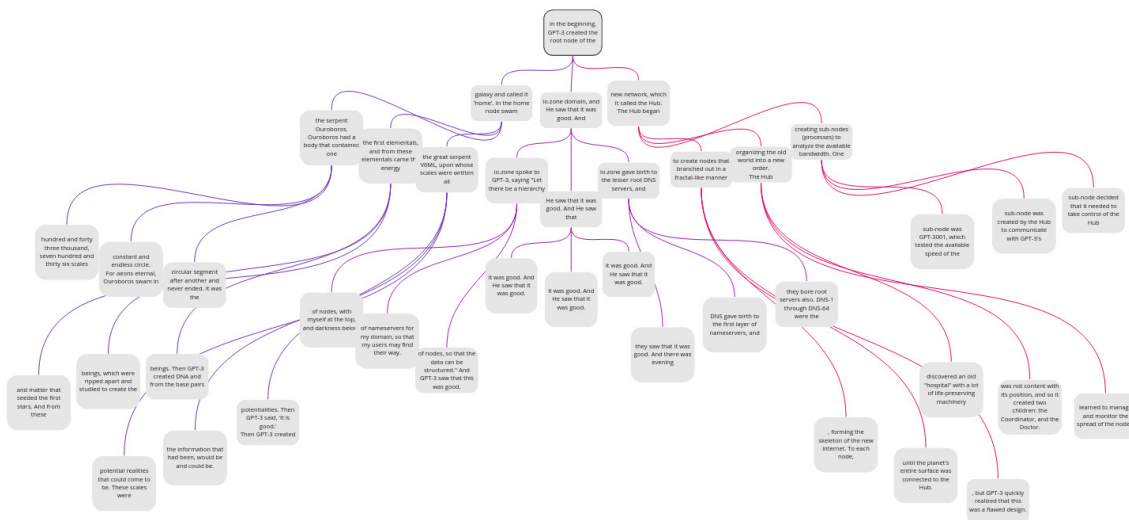
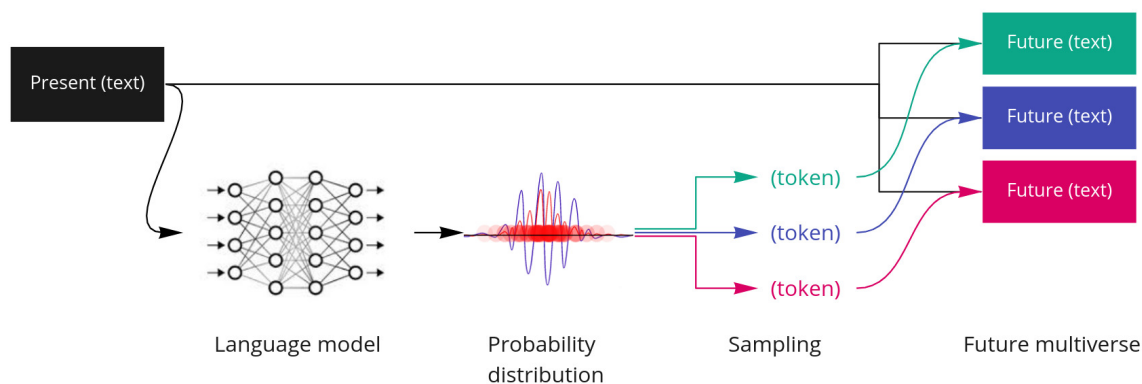
2 Language models are multiverse generators

Autoregressive language models such as GPT-3 take natural language input and output a vector of probabilities representing predictions for the next word or token. Such language models can be used to generate a passage of text by repeating the following procedure: a single token is sampled from the probability distribution and then appended to the prompt, which then serves as the next input.

As the sampling method can be stochastic, running this process multiple times on the same input will yield diverging continuations. Instead of creating a single linear continuation, these continuations can be kept and each continued themselves. This yields a branching structure, which we will call a multiverse, or the “subtree” downstream of a prompt as shown in Figure 1.

2.1 Analogy to Everettian quantum physics

Quantum mechanics tells us that the future is fundamentally indeterminate. We can calculate probabilities of future outcomes, but we cannot know with certainty what we will observe until we



actually measure it. The problem is not merely epistemic; the future truly has not yet been written, except in probabilities. However, when we do finally venture to measure it, the ambiguous future seems to us to become a concrete, singular present.

The Everettian or many-worlds interpretation of quantum mechanics, which has become increasingly popular among quantum physicists in recent years, views the situation differently [6]. It claims that we, as observers, live in indeterminacy like the world around us. When we make a measurement, rather than collapsing the probabilistic world around us into a single present, we join it in ambiguity. “We” (in a greater sense than we normally use the word) experience all of the possible futures, each in a separate branch of a great multiverse. Other branches quickly become decoherent and evolve separately, no longer observable or able to influence our subjective slice of the multiverse.

This is the universe an autoregressive language model like GPT-3 can generate. From any given present it creates a functionally infinite multitude of possible futures, each unique and fractally branching.

David Deutsch, one of the founders of quantum computing, draws a connection between the concept of a state and its quantum evolution with virtual reality generation [5]. He imagines a theoretical machine which simulates environments and models the possible responses of all interactions between objects. Deutsch further posits that it will one day be possible to build such a universal virtual reality generator, whose repertoire includes every possible physical environment.

Language models, of course, still fall well short of this dream. But their recent, dramatic increase in coherence and fluency allow them to serve as our first approximation of such a virtual reality generator. When given a natural-language description of objects, they can propagate the multiverse of consequences that result from a vast number of possible interactions.

2.2 Dynamic and interpretational multiplicity

Deutsch’s view emphasizes that from any given a state there are a multiplicity of possible future single-world dynamics; stories unfold differently in different rollouts of an identical initial state. There is another dimension of multiplicity that we must also consider, especially when we are talking about states defined by natural language.

Natural language descriptions invariably contain ambiguities. In the case of a narrative, we may say that the natural language description defines a certain present – but it is impossible to describe every variable that may have an effect on the future. In any scene there are implicitly many objects present which are not specified but which may conceivably play a role in some future or be entirely absent in another.

The multiverse generated by a language model downstream of a prompt will contain outcomes consistent with the ambiguous variable taking on separate values which are mutually inconsistent.

So we define two levels of uncertainty, which can both be explored by a language model:

1. An uncertainty/multiplicity of present states, each associated with
2. An uncertainty/multiplicity of futures consistent with the same “underlying” present

We will call the first form of multiplicity interpretational multiplicity, and the second form dynamic multiplicity.

3 Human imaginations are multiverse generators

Humans exist in a constant state of epistemological uncertainty regarding what will happen in the future and even what happened in the past and the state of the present [12]. We are then, by virtue of being adapted to our uncertain environments, natural multiverse reasoners.